RASST: Fast Cross-modal Retrieval-Augmented Simultaneous Speech Translation (maybe InfiniSST-RAG?)

First Author
Affiliation / Address line 1
Affiliation / Address line 2
Affiliation / Address line 3
`email@domain`

Second Author
Affiliation / Address line 1
Affiliation / Address line 2
Affiliation / Address line 3
`email@domain`

Abstract

Simultaneous speech translation (SST) produces target text incrementally from partial speech input. Recent speech large language models (Speech LLMs) have substantially improved SST quality, yet they still struggle to correctly translate rare and domain-specific terminology. While retrieval augmentation has been effective for terminology translation in machine translation, bringing retrieval to SST is non-trivial: it requires fast and accurate cross-modal (speech-to-text) retrieval under partial, continually arriving input, and the model must decide whether and when to apply retrieved terms during incremental generation. We propose Retrieval-Augmented Simultaneous Speech Translation (RASST), which tightly integrates cross-modal retrieval into the SST pipeline. RASST trains a lightweight speech–text retriever and performs efficient sliding-window retrieval, providing chunkwise terminology hints to the Speech LLM. We further synthesize training data that teaches the Speech LLM to leverage retrieved terms precisely. Experiments on three language directions of the ACL 60/60 dev set show that RASST improves terminology translation accuracy by up to 16% and increases overall translation quality by up to 3 BLEU points, with ablations confirming the contribution of each component.

1 Introduction

Simultaneous Speech Translation (SST) aims to generate translations incrementally as partial speech input is received. It has wide applications in international conferences and cross-lingual conversations (??). Recently, Large Language Models (LLMs) have shown strong potential as backbones for SST systems (????). However, existing SST systems struggle to accurately translate domain-specific terminologies, including technical jargon, person and location names, and abbreviations. As shown in Table ??, the terminology "masked language model" is incorrectly translated to "大规模语言模型", meaning "massive language models", potentially due to recognition error.

Source Speech: More recent work has used masked language models to fill in masked portions of the text to change labels.	
Reference: 最近的工作使用掩码语言模型来填充文本的掩码部分以更改标签。	
Plain SST: 近期的研究则使用大规模语言模型来填补文本中被遮蔽的部分，以改变标签。	
RASST: 近期的研究使用了掩码语言模型来填充文本中被掩码的部分，以改变标签。	

Table 1: A case showing plain SST without retrieval fails to translate terminology correctly while SST with retrieval is able to do so. English (green), correct Chinese terms (blue), and incorrect ones (red).

Human simultaneous interpreters often rely on external glossaries to translate specialized terminology in real time (?), and retrieval augmentation has been widely explored in machine translation (MT) to improve terminology handling (?????). Nevertheless, retrieval-augmented SST remains underexplored and introduces unique challenges (?). First, retrieval must bridge different modalities, matching speech input to entries in a text glossary. Second, retrieval must be both fast and accurate under streaming constraints. Third, since translation typically lags behind the source speech, the model must decide not only whether to use retrieved terms, but also when to apply them during incremental generation.

To address these challenges, we propose Retrieval-Augmented Simultaneous Speech Translation (RASST), an effective approach for cross-modal glossary retrieval from streaming speech and terminology-aware incremental translation. RASST trains a lightweight cross-modal speech-text retriever over short audio contexts using a contrastive loss. To keep inference efficient, we perform sliding-window retrieval that maintains high recall while adding minimal runtime cost. We further synthesize training data that teaches the Speech LLM to select relevant retrieved terms and decide when to introduce them during generation, conditioned on the prior speech context and previously generated translation. Experiments on the ACL 60/60 dev set show that RASST improves BLEU by up to 3 points on three language directions (En→Zh/De/Ja) and increases terminology translation accuracy by up to 16%, while adding at most 16% computational overhead.

2 Related Works

SST with LLM Recent work shows that simultaneous speech translation (SST) can achieve markedly higher translation quality when large language models (LLMs) serve as the backbone (?). ? adapt an LLM to SST by finetuning on a small set of synthetic translation trajectories. ? further scale this idea with an interleaving architecture that supports efficient inference over unbounded speech streams. ? study similar architectures and propose alternative training strategies and data-synthesis pipelines. ? propose a unified model for streaming transcription and translation, introducing a truncation mechanism that prunes historical speech and text based on automatically transcribed inputs. However, most existing SST systems still struggle with domain-specific terminology even using LLMs, whereas RASST integrates a terminology retriever into the SST process to translate specialized terms more accurately and reliably.

Retrieval-Augmented Translation Translating rare words and domain-specific terminology remains a persistent challenge for modern translation systems. A broad line of work addresses this via non-parametric retrieval at inference time: kNN-MT augments neural MT with nearest-neighbor lookup over a large datastore to improve domain adaptation and rare-token translation (?), and follow-up methods improve robustness by learning kernel-smoothed MT with retrieval (?) as well as improving efficiency (e.g., faster and memory-efficient retrieval pipelines) (??). In parallel, recent LLM-based MT studies investigate incorporating domain knowledge at inference time through retrieval (demonstrations or terminologies) (?). Closer to speech translation, ? propose a retrieval-and-demonstration framework that prepends retrieved speech segments as demonstrations for offline ST, and ? introduce sliding-window retrieval to localize terminology occurrences and inject retrieved terms into a speech-LLM prompt. However, these approaches are primarily designed for offline translation, whereas RASST targets streaming translation and integrates retrieval directly into the incremental translation process.

Contextual-Biasing ASR The recognition of rare words and domain-specific terminology, often referred to as contextual biasing, has also been extensively studied in automatic speech recognition (ASR). Recent work explores injecting bias lists directly into Speech LLM prompts (??). Contextual-biasing has also been investigated in streaming settings (?), but not with speech LLMs.

3 Retrieval-Augmented Simultaneous Speech Translation

We propose RASST, a retrieval-augmented simultaneous speech translation framework for terminology-sensitive streaming translation. The system has two components: a cross-modal term retriever that finds glossary entries supported by the recent speech signal, and a streaming Speech LLM that conditions on the speech stream and the retrieved term map when generating partial translations. The retriever is optimized for high-recall terminology detection under latency constraints, while the Speech LLM is trained to verify retrieved evidence against the audio and avoid unsupported terms.

3.1 Problem Formulation

Let $\mathbf{s}$ denote the source speech waveform and let $\mathcal{G} = \{(e_j, \tau_j)\}_{j=1}^{|\mathcal{G}|}$ be a bilingual glossary, where e_j is a source-language term and τ_j is its target-language translation. A simultaneous speech translation system observes $\mathbf{s}$ incrementally as a sequence of chunks $\mathbf{c}_1, \dots, \mathbf{c}_T$, where $\mathbf{c}_i = \mathbf{s}_{[\ell_i, r_i]}$ is the newly available audio between times ℓ_i and r_i. At step i, the policy has access to the speech prefix $\mathbf{s}_{\le r_i}$ and the previously emitted target prefix $\hat{\mathbf{y}}_{<i}$, and it emits a possibly empty target segment $\hat{\mathbf{y}}_i$. Empty segments correspond to wait decisions; non-empty segments are appended to the final hypothesis $\hat{\mathbf{y}}$ with delay r_i.

To ground terminology, we introduce a retriever R_ϕ that returns a step-local subset $\hat{\mathcal{G}}_i \subseteq \mathcal{G}$. The generator then models

$$p_\theta(\hat{\mathbf{y}}_i \mid \mathbf{s}_{\le r_i}, \hat{\mathbf{y}}_{<i}, \mathrm{format}(\hat{\mathcal{G}}_i)). \qquad (1)$$

The retrieved term map is evidence, not a hard constraint: the generator should use a term only when the audio context supports it.

3.2 RASST Overview

RASST runs retrieval at the same cadence as the streaming translator. For the current chunk $\mathbf{c}_i = \mathbf{s}_{[\ell_i, r_i]}$, the online retriever encodes an acoustic context that includes the current chunk plus a fixed look-back:

$$\mathbf{a}_i = \mathbf{s}_{[\max(0, \ell_i - L), r_i]}, \qquad L = 1.92\text{s}. \quad (2)$$

The look-back reduces missed terms that cross chunk boundaries. To prevent stale evidence from being injected, the online agent keeps only MaxSim evidence windows that overlap the current chunk $[\ell_i, r_i]$; a term whose best evidence lies entirely in the look-back region is not used for the current step.

The retained candidates are thresholded and capped:

$$\hat{\mathcal{G}}_i = \mathrm{TopK}_{K,\tau}\left(\{(e_j, \tau_j, R_\phi(\mathbf{a}_i, e_j))\}_{j=1}^{|\mathcal{G}|}\right), \quad (3)$$

where the main system uses $K = 10$ and a development-selected threshold $\tau = 0.78$. The resulting entries are formatted as a plain term map and supplied with the current speech chunk to the Speech LLM.

3.3 Dense Cross-Modal Term Retriever

Encoders. The retriever is a dual encoder. The text encoder E_t maps each source-language glossary term e_j to a dense embedding. The main configuration uses BGE-M3 (?) with CLS pooling and LoRA adaptation. The speech encoder E_a uses the Qwen3-Omni Audio Transformer (?), followed by a projection to the same embedding dimension. Both text and speech embeddings are ℓ_2-normalized, so every embedding has unit norm and inner product scoring is cosine similarity:

$$\mathbf{g}_j = \frac{E_t(e_j)}{\|E_t(e_j)\|_2} \in \mathbb{R}^{1024}. \quad (4)$$

Multi-scale speech windows. Terminology spans vary in duration and rarely align exactly with streaming chunk boundaries. Instead of representing the full acoustic context with a single pooled vector, the retriever creates a set of local speech-window embeddings. Given projected speech states $H = [\mathbf{h}_1, \ldots, \mathbf{h}_M]$, we average-pool over sliding encoder-frame windows:

$$\mathbf{z}_{u,v} = \frac{\frac{1}{v-u+1}\sum_{m=u}^{v}\mathbf{h}_m}{\left\|\frac{1}{v-u+1}\sum_{m=u}^{v}\mathbf{h}_m\right\|_2}. \quad (5)$$

The main retriever uses window sizes $\{2, 3, 4, 5, 6, 7, 8, 10, 12, 16, 20, 24\}$ encoder frames with stride 2 frames. With the current audio encoder frame rate, these windows cover approximately 0.16–1.92 seconds with a 0.16 second stride. This is the MaxSim retriever used in the main system; direct dense pooling over the whole audio chunk is treated as an ablation.

MaxSim scoring. For each glossary term, the retriever takes the maximum similarity over local speech windows:

$$R_\phi(\mathbf{a}_i, e_j) = \max_{\mathbf{z} \in \mathcal{Z}(\mathbf{a}_i)} \mathbf{z}^\top \mathbf{g}_j, \quad (6)$$

where $\mathcal{Z}(\mathbf{a}_i)$ is the set of multi-scale embeddings extracted from $\mathbf{a}_i$. MaxSim allows one retrieval context to contain multiple possible terms while letting each term align to its own local acoustic evidence. In implementation, text embeddings are precomputed once; online retrieval scores the current set of speech-window embeddings against the active glossary bank, then applies top-K and threshold filtering.

3.4 Retriever Training Data

We train the retriever from GigaSpeech-derived speech-text supervision and additional Wiki-synthetic terminology examples. For the speech-derived data, we first obtain word-level timestamps with the Montreal Forced Aligner (MFA), extract terminology candidates from transcripts, normalize their surface forms, and map each candidate to its timestamp span. Exact repeated MFA term events are deduplicated. For each selected acoustic context, every known MFA term event overlapping that context is written as a positive.

The main training data uses variable-duration contexts of 2.88, 3.84, 4.80, and 5.76 seconds. These durations match the online retrieval contexts induced by latency multipliers 1–4 plus the fixed 1.92 second look-back. The Wiki-synthetic data broadens term coverage beyond terms observed in GigaSpeech transcripts and is ablated separately from the GigaSpeech-only setting.

3.5 Retriever Objective

We train with a multi-positive contrastive objective. For a speech example b, let $\mathcal{P}_b$ be the set of positive terms and let $\mathcal{C}_b$ contain positives, in-batch negatives, and mined hard negatives. The loss is

$$\mathcal{L}_{\text{ret}} = -\log \frac{\sum_{p \in \mathcal{P}_b} \exp(R_\phi(\mathbf{a}_b, p)/\gamma)}{\sum_{c \in \mathcal{C}_b} \exp(R_\phi(\mathbf{a}_b, c)/\gamma)}, \quad (7)$$

where $\gamma = 0.07$ is the contrastive temperature. The main retriever uses hard-negative depth HN1024, i.e., up to 1024 mined hard negatives per sample. We fine-tune both encoders with LoRA (?); the main retriever uses rank 128 and alpha 256 for both the Qwen3-Omni audio encoder and the BGE-M3 text encoder.

MFA-supervised MaxSim. Because global MaxSim can match a positive term to any local window in the context, we use MFA timestamps to localize positive supervision. For a positive term with timestamp span $[b_p, e_p]$, we restrict its positive score to speech windows that cover the span:

$$\mathcal{Z}_p = \{\mathbf{z}_{u,v} : u \le b_p,\ v \ge e_p\}. \quad (8)$$

The main setting selects the shortest covering window for each positive term, encouraging tight acoustic alignment. If no window covers the term, or if timestamps are unavailable, the training example falls back to standard global MaxSim.

3.6 Online Retrieval

At inference time, we pre-encode the glossary terms and build a dense inner-product retrieval bank. For each streaming step, the HN1024 retriever encodes the current chunk plus the 1.92 second look-back, computes MaxSim scores, removes evidence windows that do not overlap the current chunk, and retains the top 10 candidates above $\tau = 0.78$. Duplicate candidates from overlapping evidence windows are merged by keeping the maximum score. The selected entries are formatted as

$$\texttt{source\_term} \mapsto \texttt{target\_translation} \quad (9)$$

and injected into the Speech LLM state for that step. The threshold $\tau = 0.78$ is selected from development-only fixed-raw-denominator precision-recall calibration over raw, 10k, and 100k glossary banks.

3.7 Retrieval-Augmented SST Generator

The generator follows the InfiniSST streaming formulation (?) and uses Qwen3-Omni-30B-A3B-Instruct as the Speech LLM. At each step, the model receives the speech history, the previous target prefix, and the current term map. The map is not enforced by decoding constraints; the model must decide whether each entry is supported by the audio.

We construct term-map SFT data to make this behavior match runtime retrieval. Starting from InfiniSST-style speech-translation conversations, we identify ground-truth terms by source-side string matching against a 100k glossary and keep only terms whose target translations are supported by the assistant target text. For chunks with ground-truth terms, we run the retriever on the speech chunk to add acoustically plausible negative terms, then cap the term map at 20 entries. We further augment target translations with LLM-proposed variants that preserve rare terminology while varying common synonyms. Chunks without supported ground-truth terms are assigned an empty term map. Finally, matching assistant output spans are wrapped with <term>...</term> tags, which are stripped before evaluation. We fine-tune the Speech LLM with LoRA rank 32 and alpha 64 using standard supervised cross-entropy on assistant tokens.

3.8 Evaluation Protocol

We evaluate the retriever offline for threshold selection and the full system online for simultaneous translation. Retriever thresholds are selected only on development data using fixed-raw-denominator precision-recall curves; ACL and medicine results are not used to tune τ, checkpoints, or hyperparameters. End-to-end evaluation reports StreamLAAL for latency, SacreBLEU for translation quality, and TERM_ACC as the main terminology metric. TERM_ACC uses a fixed raw-glossary denominator: expanded retrieval banks can change which terms are retrieved at runtime, but they do not expand the set of terms counted for the main accuracy numerator or denominator.

4 Experimental Setups

4.1 Data

Retriever Training We synthesize the retriever training data from GigaSpeech (?) and Wikidata[1]. GigaSpeech is a multi-domain English ASR

[1] https://www.wikidata.org/

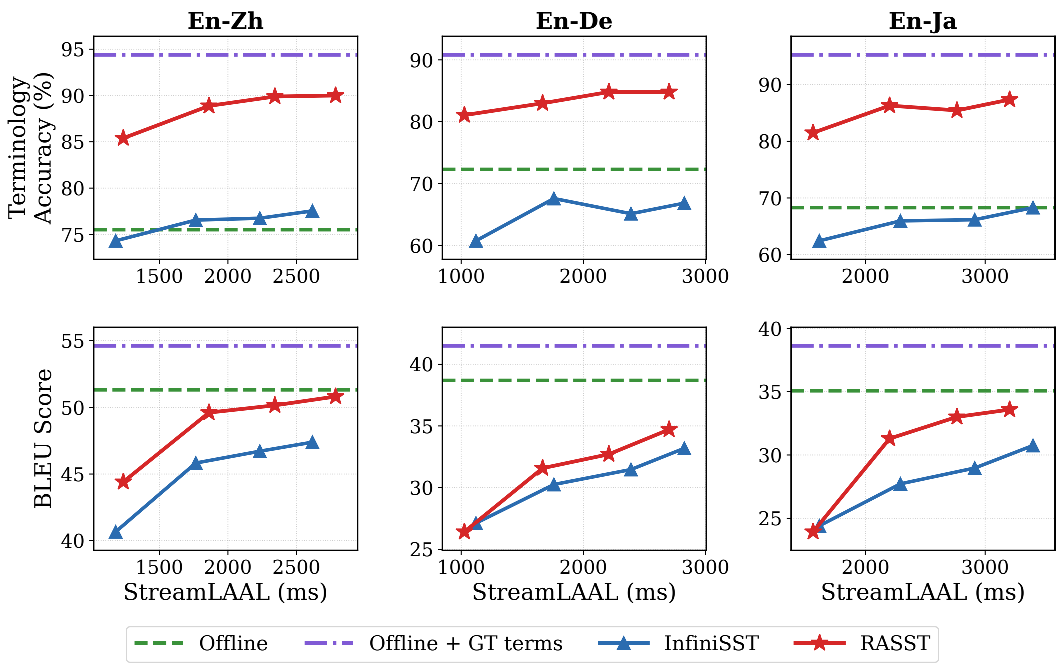

Figure 1: Quality–latency trade-off of RASST and baselines on the ACL 60/60 dev set. Quality is measured by both terminology accuracy and BLEU. Across all three language directions, RASST improves terminology accuracy over InfiniSST by 10% while also achieving higher translation quality.

corpus containing 10K hours of high-quality labeled audio. We process it following the procedure in Section ??, yielding 4.7M training pairs. The variable speech-context duration is sampled uniformly from $2.84, 3.96, 4.80, 5.76$ seconds, corresponding to the inference-time chunk sizes $0.96, 1.92, 2.84, 3.84$ plus a $t_{\text{lb}} = 1.92$-second look-back. The look-back captures terms that cross chunk boundaries and we set it to 1.92 seconds since most terms are shorter than this, as shown in Figure ??.

For Wikidata, we extract candidate terms from the Wikidata RDF truthy dump using entities that have both an English `rdfs:label` and at least one `wdt:P31` instance-of triple. We normalize each label by removing parenthetical qualifiers, lowercasing for deduplication, and retaining only short alphabetic or hyphenated terms with at most five words. We further require a non-empty short description from the glossary index that does not simply repeat the term, which filters out many ambiguous or non-English labels. To avoid a corpus dominated by very frequent instance types such as people, taxa, or scholarly articles, we assign each entity a primary `P31` type using its least frequent instance type and rank terms with inverse-type-frequency sampling. Terms appearing in the held-out evaluation glossaries are removed before training. For the resulting top-ranked terms, Gemini-2.5-Flash generates three natural 5–15-word utterance variants that contain the term verbatim using the prompt template in Appendix ??; generations that fail this constraint are discarded. We synthesize the utterances with the Fun-CosyVoice3-0.5B-2512 checkpoint of CosyVoice 3, using GigaSpeech speaker prompts.

Speech LLM Training We synthesize the Speech LLM training data using the same GigaSpeech ASR dataset. The interleaved En-Zh/De/Ja translation data is synthesized similar to InfiniSST with details in Appendix ??. We subsample 12.5K instances for Speech LLM fine-tuning. We build a 100k glossary for distractor retrieval of G_i and we cap the size of G_i at 20.

Evaluation We evaluate on two datasets. The first is the ACL 60/60 dev set (?), which contains five English ACL talks translated into 10 target languages. We evaluate on the En-Zh/De/Ja directions, feeding the unsegmented talks directly to the model. This dataset includes human-annotated terminology, which we use to construct the glossary for the model to retrieve from. After deduplication, the glossary contains 238 unique terms. The second is the ESO test set (?), which consists of five long-form pre-recorded English oncology training videos provided by the European School of Oncology, with translations into French, Spanish, German, and Slovene. We use GPT-5.4[2] to supplement the dataset with Chinese and Japanese translations, and to extract a candidate terminology list. After manual verification, the resulting glossary contains 217 terms. To test robustness under larger glossaries, we extend both glossaries to 1K and 10K terms with terms drawn from Wikidata that are unseen during training.

4.2 Evaluation Metrics

Latency Following IWSLT 2025 practice (?), we use StreamLAAL (?) as our latency metric. StreamLAAL segments the model hypothesis to align with reference sentences and reports the average latency over the resulting (hypothesis, reference) pairs.

Translation Quality We evaluate translation quality on the aligned (hypothesis, reference) sentence pairs using SacreBLEU (??).

Terminology Accuracy We introduce terminology accuracy to measure how accurately the model translates terms. For each terminology occurrence in the reference, we check whether its translation appears as an exact match in the aligned hypothesis. The metric is the percentage of such occurrences correctly matched, ranging from 0 to 100%.

4.3 Model Configuration

Architecture For the retriever, we use a Qwen3-Omni Audio Transformer[3] as the speech encoder, followed by a linear projection. For text encoding, we use the dense BGE-M3[4] encoder and extract the feature of [CLS] token. For the Speech LLM, we use Qwen3-Omni-30B-A3B-Instruct[5].

Training We train both the cross-modal retriever and the Speech LLM with LoRA (?). For retriever training, the number of hard negative N is 1024 and the global batch size is 8192. For Speech LLM training, we use sequence packing and a global batch size of 8192 tokens. Full hyperparameters and details are provided in Appendix ??.

[2] https://developers.openai.com/api/docs/models/gpt-5.4

[3] https://huggingface.co/Atotti/Qwen3-Omni-AudioTransformer

[4] https://huggingface.co/BAAI/bge-m3

[5] https://huggingface.co/Qwen/Qwen3-Omni-30B-A3B-Instruct

Inference We implement the inference agent in SimulEval (?), varying the chunk size l over $0.96, 1.92, 2.88, 3.84$ seconds. The Speech LLM is invoked once per chunk and generates up to 40 new tokens per call. We serve it with vLLM (?) using tensor parallelism of size 2, decode with temperature 0.6, top-p=0.95, and top-k=20, and strip the assistant's `<term>` tags from the hypothesis before scoring.

For retrieval, we build a FAISS inner-product index over ℓ_2-normalized term embeddings. At each step, the retriever encodes the current speech chunk together with a 1.92-second look-back, and queries the index using only those windows whose right boundary lies within the current chunk. We use Top-K=10 and discard candidates with similarity below τ=0.78[6].

4.4 Baselines

Offline We use Qwen3-Omni-30B-A3B-Instruct to translate pre-segmented speech utterances offline, with the same decoding parameters as RASST. This serves as an upper bound on translation quality.

Offline + GT Terms We extend the Offline baseline by adding the ground-truth terms appearing in each utterance to the prompt in a manner similar to ?, helping the model handle terminology correctly. This serves as an upper bound on terminology translation accuracy.

InfiniSST InfiniSST (?) is a strong SST baseline for simultaneous translation of unbounded speech and achieved top performance in the low-latency track at IWSLT 2025 (??). We train InfiniSST on the same data and with the same Speech LLM as RASST, but remove all terminology input G_i. This setting can be viewed as RASST without retrieval.

5 Results and Analysis

5.1 Main Results

RASST substantially improves terminology translation accuracy As shown in the first row of Figures ?? and ??, RASST consistently improves terminology translation accuracy over the strong InfiniSST baseline on the ACL tagged raw glossary and on the medicine hardraw readout. The horizontal offline references separate full-context decoding from full-context decoding with ground-truth

[6] We select τ on the dev set to allow at most a 1% drop in recall.

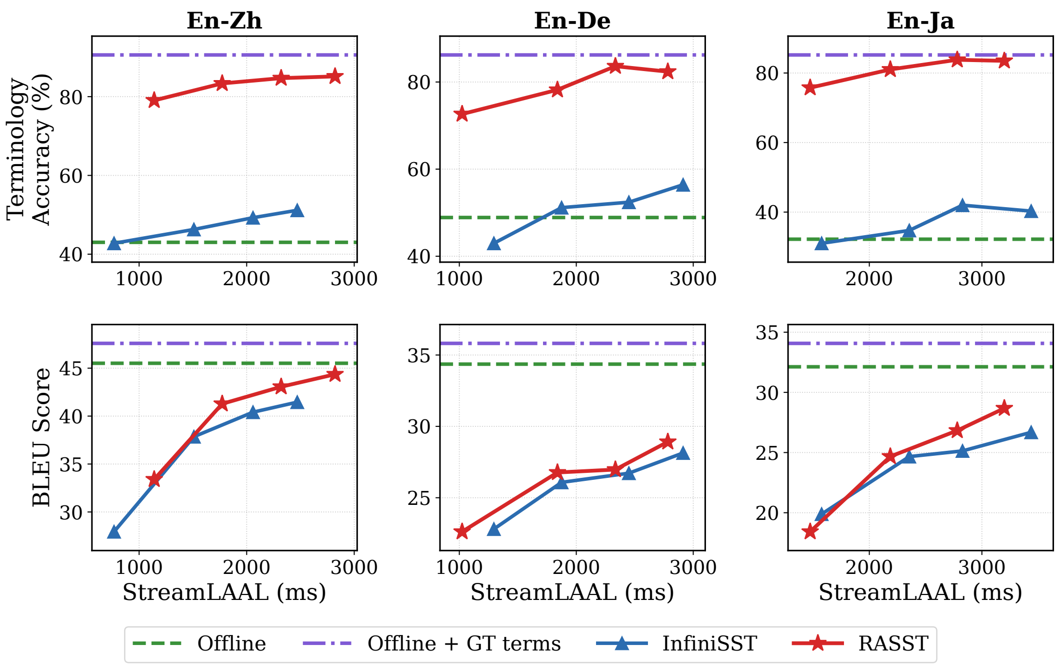

Figure 2: Quality–latency trade-off of RASST and baselines on the ESO test set. Quality is measured by both terminology accuracy and BLEU. Across all three language directions, RASST improves terminology accuracy over InfiniSST by 30% while also achieving slightly higher translation quality.

glossary terms, showing the behavior of the same offline LLM when oracle terminology is supplied. The ACL tagged results cover En-Zh, En-De, and En-Ja under the fixed raw-glossary denominator, while the medicine hardraw panel is a manually curated hard-domain stress test whose denominator emphasizes specialized clinical terms and exact target forms that are difficult for a retrieval-free streaming system.

RASST improves overall translation quality As shown in the second row of Figures ?? and ??, RASST also improves overall translation quality in the strongest ACL tagged and clean medicine En→Zh comparisons. The BLEU gains are smaller than the terminology-accuracy gains, reflecting that specialized terms occur relatively infrequently in the speech and that the main effect of retrieval is concentrated on those terms.

Computational efficiency Figure ?? reports the retriever-side compute budget under the same timeline retrieval policy used by RASST. Since retrieval is placed on a separate GPU and is invoked once per vLLM call, this is not additional user-visible StreamLAAL by itself; it measures how much retriever compute must fit inside each generation cadence interval. The median retrieval call stays around 37–44 ms, and normalizing by the slower LM cadence makes the RAG compute share decrease as StreamLAAL increases.

5.2 Ablation Studies

We group ablations by the component they test: retriever data and architecture, retriever calibration, Speech LLM term-map supervision, and online retrieval behavior. Unless otherwise stated, all terminology metrics use the same fixed raw-glossary denominator as the main results.

Retriever training data Figure ?? compares the main retriever against a GigaSpeech-only variant using the same development readout used for main retriever selection. Adding Wiki-synthetic terminology supervision improves dev Recall@10 with the GS-10k bank from 98.30 to 99.01. The difference is larger under stricter score filtering: at τ=0.80, filtered dev recall increases from 95.86 to 97.91. This suggests that the synthetic text-derived terminology examples mainly improve robust retrieval under noisy or expanded runtime banks, rather than only increasing unfiltered top-k recall.

Retriever context construction Table ?? compares fixed-context retrievers with the variable-context training used by RASST. The fixed 5.76s setting is strongest because it always exposes the longest acoustic span. The variable-context retriever is

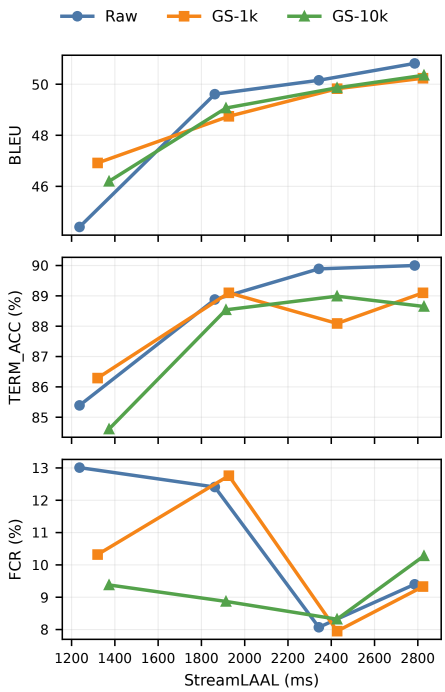

Figure 3: Scaling the glossary from 0.2K to 1K and 10K terms. RASST shows only a slight drop in terminology accuracy and translation quality, remaining well above the no-retrieval baseline.

Training context	Raw	GS-10k	GS-100k
Fixed 1.92s	98.11	97.47	95.18
Fixed 3.84s	98.65	98.31	97.58
Fixed 5.76s	99.64	99.58	99.33
Variable 2.88–5.76s	99.20	98.97	98.58

Table 2: Retriever context-duration ablation on development data. Values are Recall@10 (%) under fixed raw-denominator scoring. Raw uses only the per-context development glossary as the retrieval bank, while GS-10k and GS-100k add GigaSpeech-derived distractor terms to the runtime bank without changing the scoring denominator.

within 0.44, 0.61, and 0.75 recall points of fixed 5.76s on the raw, GS-10k, and GS-100k banks, respectively, while matching the online contexts induced by latency multipliers 1–4 plus the fixed 1.92s look-back. We therefore use variable-context training in the main system: it retains most of the long-context retrieval benefit without assuming that every streaming step has a fixed 5.76s context.

Retriever encoders Figure ?? tests whether the main Qwen3-Omni-AuT audio encoder and BGE-M3 dense text encoder are necessary. The GigaSpeech-expanded banks increase only the retrieval candidate set, so a robust encoder should preserve high recall even when many extra distractor terms are present. Replacing BGE-M3 with multilingual-E5 hurts robustness as the bank expands, while replacing Qwen3-Omni-AuT with WavLM has a smaller but consistent cost under the same fixed-denominator protocol.

Hard negatives and threshold sweep Figure ?? shows the available development-only evidence for hard-negative training and threshold selection. We use the fixed raw-glossary denominator when sweeping τ so that larger retrieval banks affect only the retrieved candidates, not the evaluation scope. HN1024 gives the strongest precision-recall trade-off among the available curves. The dashed line indicates the recall-preserving operating point: the largest threshold before the HN1024 curve has lost 1 percentage point of recall relative to its raw-bank reference, which places the selected deployment threshold near τ=0.78.

Speech LLM term-map supervision Figure ?? isolates the Speech LLM term-map supervision effect on En-De. The no-TM-SFT system uses runtime RAG without term-map supervision, while the historical LLM-generated TM-SFT line trains the Speech LLM on dense LLM-generated term maps. The current RASST line replaces those dense LLM-generated maps with the retriever-style term maps used in the main system, improving terminology accuracy while moving toward the oracle-term upper bound.

Runtime glossary-bank expansion Figure ?? tests whether RASST remains stable as the runtime terminology bank grows. The evaluation denominator is fixed to the raw tagged glossary, so changes in TERM_ACC reflect retrieval and generation behavior under additional distractor terms rather than a changing metric scope. Expanding to GS-1k and GS-10k preserves BLEU and keeps TERM_ACC close to the raw-bank setting across latency multipliers, while FCR remains controlled. This shows that the system remains usable with larger runtime banks, though the appendix stress test shows that bank quality and duplicate control still matter. Full ACL rows and the stricter medicine stress-test rows are reported in Appendix ??.

MaxSim inference Figure ?? examines the contribution of multi-scale inference. The main retriever scores candidate terms with MaxSim over speech-window embeddings, allowing short terminology mentions to match the most relevant part of the current speech context. Reusing the same retriever but

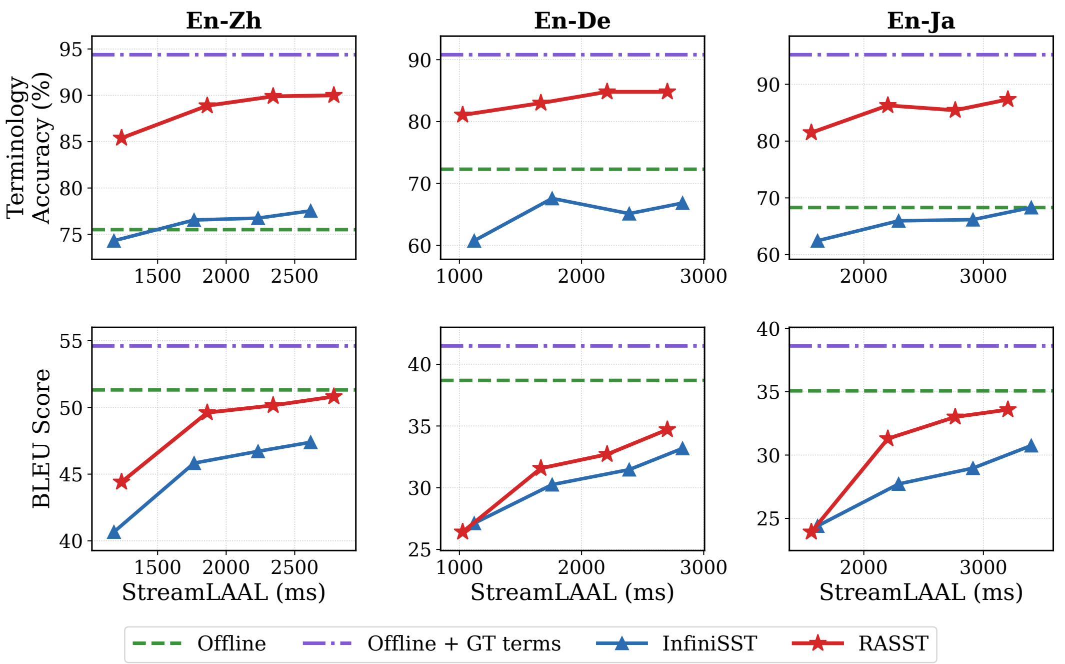

Figure 4: Main results on the ACL tagged raw glossary with the final HN1024 retriever at τ=0.78. We compare Terminology Accuracy (top) and BLEU (bottom) against StreamLAAL for En-Zh, En-De, and En-Ja. Horizontal lines show full-context offline LLM references, without and with oracle glossary terms. RASST improves terminology accuracy over the InfiniSST baseline across all three language directions while maintaining competitive translation quality.

querying only the largest internal window lowers dev Recall@10 on the GS-100k bank from 98.58 to 96.07, and lowers filtered recall at τ=0.75 from 98.42 to 93.62. This shows that the smaller windows are not redundant at inference time. We also compare against a historical dense 1.92s retriever trained directly as a single audio embedding, which removes the MFA-window training–inference mismatch but underperforms on the expanded bank, reaching 92.65 Recall@10 and only 23.94 filtered recall at τ=0.75. The result indicates that dense one-shot speech embeddings have much weaker score calibration for thresholded retrieval than the multi-scale MaxSim formulation.

Real-time factor The RAG compute-RTF readout in Figure ?? is reported separately from quality ablations because it measures runtime budget rather than retrieval accuracy. End-to-end latency is still represented by StreamLAAL in the main result curves.

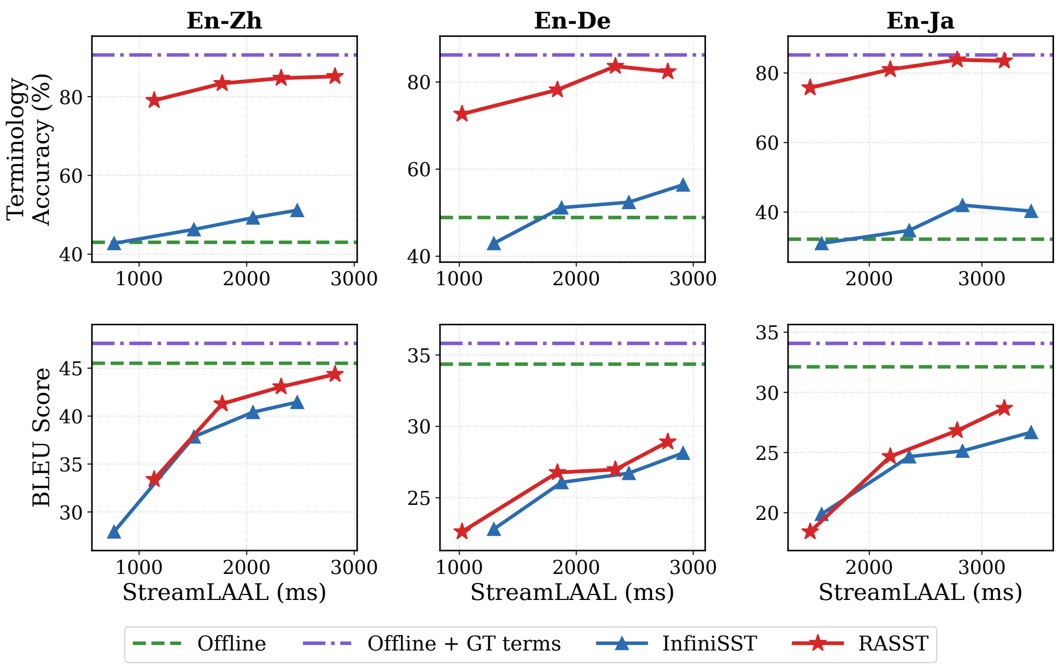

Figure 5: Main results on the manually curated medicine hardraw glossary. This is a strict domain stress test built from specialized clinical terms and exact target forms, rather than a broad easy glossary. We compare Terminology Accuracy (top) and BLEU (bottom) against StreamLAAL for En-Zh, En-De, and En-Ja. Horizontal lines show full-context offline LLM references, without and with oracle glossary terms.

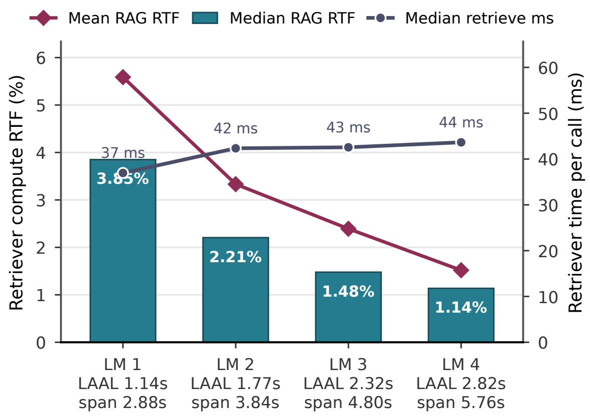

Figure 6: RAG compute real-time factor under timeline retrieval. We compute $\mathrm{RTF}_{\mathrm{RAG}} = t_{\mathrm{retrieve}}/(0.96\,\mathrm{s} \times \mathrm{LM})$ from verified En-Zh medicine hardraw runtime traces. The retriever runs once per vLLM generation step and encodes the current generation span plus a fixed $1.92\,\mathrm{s}$ look-back. Bars show median per-call RTF, the magenta line shows mean RTF, and the gray line shows median retriever time. Even as the encoded span grows from 2.88 to 5.76 seconds, the median RAG compute share falls from 3.85% to 1.14%.

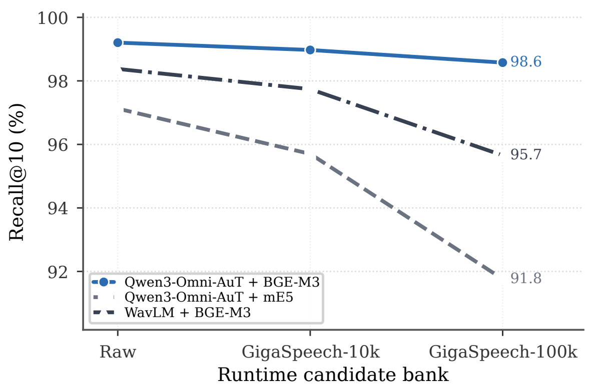

Figure 7: Retriever encoder ablation on the development set. The Qwen3-Omni-AuT + BGE-M3 line is our main retriever encoder configuration. All points report Recall@10 with the denominator fixed to the dev raw glossary. Raw uses only the development glossary as the candidate bank, while GigaSpeech-10k and GigaSpeech-100k add 10k or 100k GigaSpeech-derived distractor terms to the runtime retrieval bank without changing the scoring denominator.

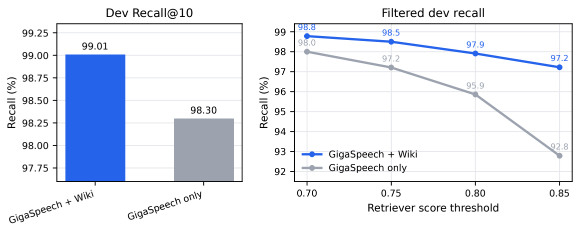

Figure 8: Retriever training-data ablation on the development set. The main retriever uses both GigaSpeech-derived supervision and Wiki-synthetic terminology examples, while the ablation removes the Wiki-synthetic rows and trains only on GigaSpeech. The left panel shows the checkpoint-selection metric, dev Recall@10 with a GS-10k retrieval bank. The right panel shows threshold-filtered dev recall with the same bank.

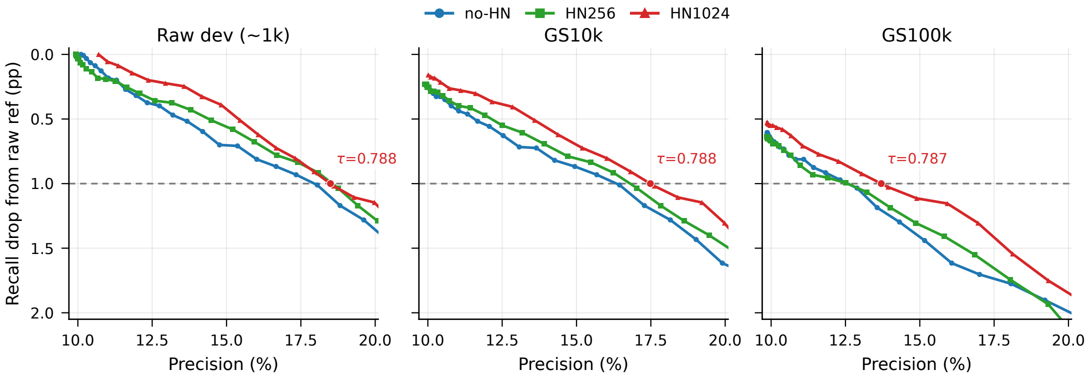

Figure 9: Retriever hard-negative and threshold calibration ablation on development data. The figure reports fixed-raw-denominator precision-recall curves over multiple retrieval-bank sizes for no-HN, HN256, and HN1024. HN1024 is the main setting. The dashed vertical line marks the dev-only threshold-selection point where HN1024 recall drops by 1 percentage point from its raw-bank reference, yielding the operating region used by RASST.

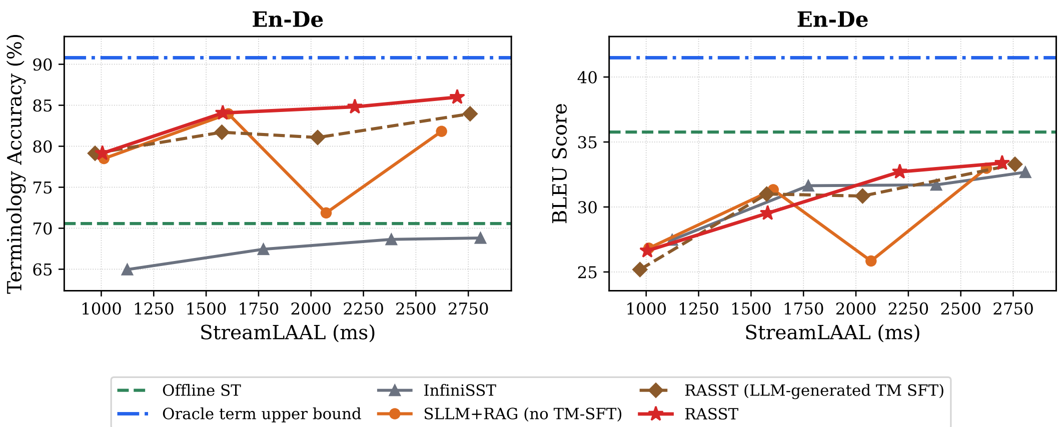

Figure 10: Speech LLM ablation on En-De ACL tagged evaluation. We reuse the En-De rows from the earlier tagged main-result figure, rename its old RASST line as RASST (LLM-generated TM SFT), and add the current RASST En-De result from Figure ??. The blue horizontal line is the full-context offline LLM upper bound with oracle glossary terms.

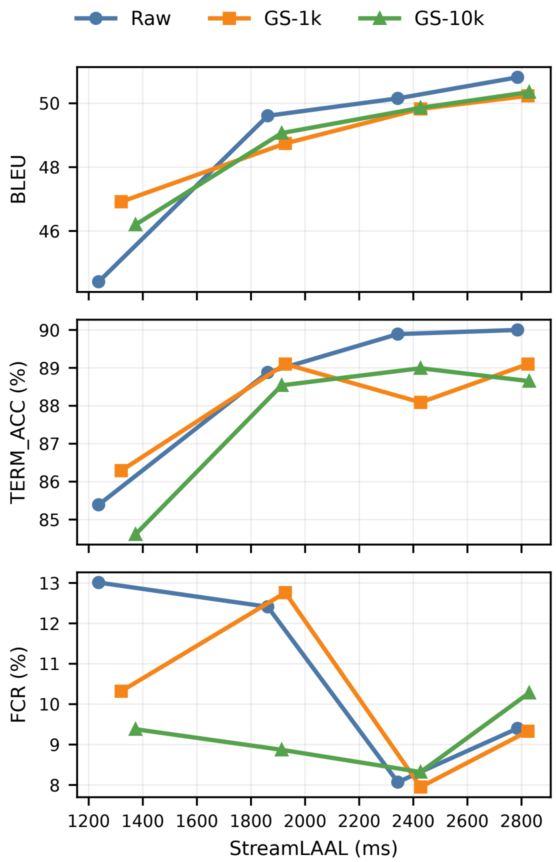

Figure 11: Runtime glossary-bank ablation on the En-Zh ACL tagged evaluation. The runtime retrieval bank changes from the raw tagged glossary to GS-1k and GS-10k expanded banks, while terminology metrics are always scored against the fixed raw tagged denominator. Points are ordered by latency multiplier from 1 to 4.

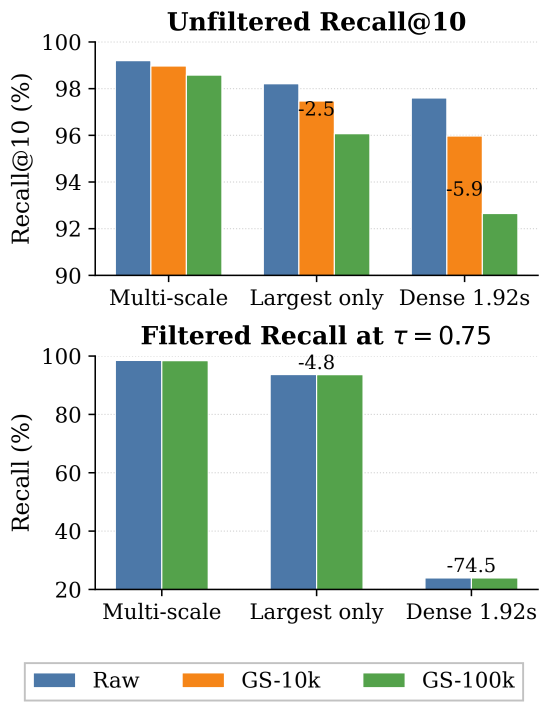

Figure 12: Multi-scale inference ablation on development data. The first two systems reuse the same MFA-supervised variable-context retriever and fixed raw-denominator scoring; only the runtime MaxSim window set changes from all windows to the largest internal window. The dense 1.92s system is the historical single-embedding retriever trained without MFA window localization. Expanded banks add GigaSpeech distractors only at retrieval time.

6 Conclusion

We presented RASST, a retrieval-augmented simultaneous speech translation framework for accurate domain-specific terminology translation. RASST integrates a lightweight cross-modal retriever with a Speech LLM, and uses noise-robust training to remain effective under retrieval errors during inference. On ACL 60/60 dev set, RASST outperforms strong baselines, improving terminology accuracy by up to 15% on En→Zh/De/Ja. RASST remains effective with glossaries automatically extracted from conference papers, supporting practical deployment without curated term lists.

Limitations

This document does not cover the content requirements for ACL or any other specific venue. Check the author instructions for information on maximum page lengths, the required "Limitations" section, and so on.

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A Speech LLM Data Synthesis

GigaSpeech does not provide translations. We therefore translate transcripts into Chinese/German/Japanese using Qwen3-30B-A3B-Instruct-2507-FP8[7] with prompt ?? and filter out low-quality translations with MetricX-QE (?)[8] by discarding instances with scores below -3.0. We subsample 12.5K instances for Speech LLM fine-tuning.

Listing 1: The prompt template used for synthesizing translation.

```
You are given English document split into lines.
     Translate each line into Chinese. Do not
    include any other text.
<begin>
{}
<end>
```

[7] https://huggingface.co/Qwen/Qwen3-30B-A3B-Instruct-2507-FP8

[8] https://huggingface.co/google/metricx-24-hybrid-xl-v2p6

B Wikidata Utterance Generation Prompt

The prompt template below is used to elicit term-anchored English utterances for Wikidata-derived retriever training examples. Each batch contains a JSON list of terms and short descriptions, and the model is required to return raw JSON.

```
System instruction:
You are a dataset builder for speech recognition
research.
Your task: given a list of terms with their short
descriptions, generate short English utterances
(5-15 words each) that a speaker might say in a
lecture, tutorial, or technical discussion. Use
the description as context to produce accurate,
meaningful sentences, but do NOT just copy the
description.
Each utterance MUST contain the given term
EXACTLY as written (same spelling, but case
may differ). Vary sentence structure and term
position (beginning, middle, end). Do NOT repeat
the same pattern across terms.
Output valid JSON only - no markdown fences, no
explanation.

User prompt:
Generate {variants_per_term} short utterances
(5-15 words) for EACH of the following terms. Use
the description as context to make the utterance
accurate and meaningful. Each utterance must
contain the term verbatim.

Terms with descriptions:
[
  {"term":    "Seismology",   "description":
"scientific study of earthquakes"},
  {"term":  "dataflow  graph",  "description":
"graph representation of data dependencies"}
]

Return a JSON object mapping each term to a
list of {variants_per_term} utterance strings.
Example format:
{
  "Seismology": [
    "Seismology helps us understand earthquake
patterns.",
    "The  field  of  seismology  has  advanced
rapidly."
  ]
}

IMPORTANT: output raw JSON only, no markdown
code fences.
```

C Term Duration Distribution

We show the distribution of term duration on GigaSpeech training set in Figure ??.

D Training Data Prompt Template

Figure ?? illustrates a complete training instance for the SST model.

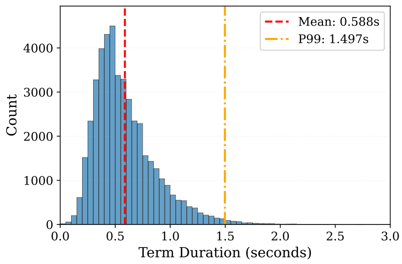

Figure 13: The distribution of term duration on GigaSpeech training set.

E Glossary Extraction Details

To simulate a realistic deployment scenario, we extract a glossary from the reference paper PDFs using `Gemini-2.5-Flash`[9].

E.1 Prompting Strategy

We feed the full text of each paper into the model and use a structured prompt to identify domain-specific terms, acronyms, and their corresponding translations in Chinese, German, and Japanese. The exact prompt template is provided in Listing ??. To ensure robustness, we re-prompts the model to fix malformed JSON outputs if parsing fails. Finally, we deduplicate terms within each paper using case-insensitive matching.

Listing 2: The prompt template used for terminology extraction and translation.

```
Extract technical glossary terms from the
     following ACL paper text and provide
     translations.

Focus on:
- Domain-specific terms (NLP / speech / ML)
- Terms likely mentioned in oral presentations
- Substantive technical concepts (not common
     words)
- Acronyms

For each term, provide:
1. The term itself (in English)
2. Translations in Chinese (zh), German (de),
     and Japanese (ja)

Return ONLY a valid JSON array (no markdown, no
     commentary). Format:
[
  {
    "term": "BERT",
    "target_translations": {
      "zh": "BERT",
      "de": "BERT",
      "ja": "BERT"
    }
  },
  {
    "term": "attention mechanism",
    "target_translations": {
      "zh": "注意力机制",
      "de": "Aufmerksamkeitsmechanismus",
      "ja": "注意機構"
    }
  }
]

Constraints:
- Extract as many relevant technical terms as
    you can.
- Do not include author names, affiliations, or
    generic phrases.
- If the same term appears multiple times, list
    it only once.
- For acronyms, provide the acronym itself (not
    the full form).

PAPER_TEXT:
__PAPER_TEXT__

Return ONLY the JSON array, nothing else.
```

[9] https://docs.cloud.google.com/vertex-ai/generative-ai/docs/models/gemini/2-5-flash

E.2 Extraction Examples

Table ?? shows a subset of terms automatically extracted from the paper "Online Semantic Parsing for Latency Reduction in Task-Oriented Dialogue" (2022.acl-long.110). The model successfully identifies specific acronyms (e.g., FLR, SimulMT) and technical concepts (e.g., online semantic parsing).

F Runtime Glossary-Bank Ablation Details

Table ?? reports the complete En-Zh runtime glossary-bank ablation. The main text plots only the ACL tagged rows to keep the ablation compact. We include the medicine rows here as a stricter curated-domain stress test: the denominator is manually verified and intentionally targets specialized clinical terms and exact target forms that are difficult for streaming systems. The rows use the same fixed raw-denominator scoring rule, but the larger expanded banks contain more near-duplicate and variant translations, which can reduce exact TERM_ACC even when BLEU remains stable.

G Training Details

Retriever. We train the retriever with LoRA applied to both the speech and text encoders. For the audio encoder, LoRA is inserted into the query/key/value/output projections and MLP blocks with rank r=32 and scaling factor α=64. For the text encoder, LoRA targets the query/key/value projections with rank r=16 and α=32. We train with a multi-positive InfoNCE objective using a fixed temperature τ=0.03. We optimize with AdamW

System	You are a professional simultaneous interpreter. You will be given chunks of English audio and you need to translate the audio into Chinese text. Use the term_map as a reference for terminology if provided.
Audio	audio_chunk_0.wav → audio_chunk_1.wav → audio_chunk_2.wav
User 1	<audio> term_map: things → 事; everything → 所有努力; right → 权利; thing → 东西; that thing → 那东西.
Assistant 1	她会做些我本意不想让她做的<term>事</term>。这样一来，之前<term>所有努力</term>都白费了，我不得不重新写一遍。
User 2	<audio> term_map: the Porch Gable → 门廊小屋; story → 传说; Anne → 安妮; seclusion → 静谧; Diana → 戴安娜; gabelle → 法国盐税; that story → 这个故事; porch → 门廊.
Assistant 2	然而，最终<term>传说</term>终于完成了，<term>安妮</term>在<term>门廊小屋</term>的<term>静谧</term>中把它读给了<term>戴安娜</term>听。她
User 3	<audio> term_map: Diana → 戴安娜; watchful eye → 密切留意; Robert Ray → 罗伯特·雷; her pathetic scene → 那个悲情场景; scene → 现场; that scene → 那场戏.
Assistant 3	成功地呈现了那个悲情场景，同时又没有牺牲<term>罗伯特·雷</term>，她一边读一边<term>密切留意</term>着<term>戴安娜</term>的反应。戴安娜

Figure 14: A visualized Chinese new_v10 training sample. Each user turn supplies one audio chunk and a retrieved term map; the assistant target is the corresponding partial translation with supervised term tags.

Extracted Term	Target Translations (Zh / De / Ja)
online semantic parsing	Zh: 在线语义解析; De: Online-Semantikparsing; Ja: オンライン意味解析
FLR	Zh: 最终延迟减少; De: Endlatenzreduzierung; Ja: 最終レイテンシ削減
SimulMT	Zh: 同声机器翻译; De: Simultane MÜ; Ja: 同時機械翻訳
ASR	Zh: 自动语音识别; De: Automatische Spracherkennung; Ja: 自動音声認識
dataflow graph	Zh: 数据流图; De: Datenflussgraph; Ja: データフローグラフ
PREFIXTOGRAPH	Zh: PREFIXTOGRAPH; De: PREFIXTOGRAPH; Ja: PREFIXTOGRAPH

Table 3: Examples of extracted terms and their multilingual translations from paper 2022.acl-long.110. Note that acronyms or specific model names (e.g., PREFIXTOGRAPH) are often preserved in target languages.

(learning rate 1×10^{-4}, weight decay 0.01), cosine annealing with 10% warmup, and gradient clipping (max norm 1.0) under BF16 mixed precision.

Dataset	Bank	LM	BLEU	StreamLAAL	TERM_ACC	RealAdopt	FCR
Tagged ACL	raw	1	44.41	1237	85.39	89.40	13.01
	raw	2	49.61	1862	88.88	90.37	12.41
	raw	3	50.15	2343	89.89	91.08	8.07
	raw	4	50.81	2786	90.00	92.12	9.40
	gs1k	1	46.92	1320	86.29	89.84	10.32
	gs1k	2	48.74	1927	89.10	90.48	12.76
	gs1k	3	49.82	2427	88.09	89.99	7.95
	gs1k	4	50.23	2824	89.10	91.36	9.33
	gs10k	1	46.20	1373	84.61	88.11	9.38
	gs10k	2	49.07	1913	88.54	90.29	8.87
	gs10k	3	49.86	2427	88.99	91.25	8.32
	gs10k	4	50.35	2828	88.65	91.37	10.28
Medicine	raw	1	33.40	1140	79.05	84.23	30.79
	raw	2	41.28	1772	83.36	90.52	27.51
	raw	3	43.06	2321	84.70	91.26	27.27
	raw	4	44.36	2822	85.14	92.15	25.96
	gs1k	1	36.37	1293	75.19	85.66	30.07
	gs1k	2	42.11	1874	78.45	87.66	23.27
	gs1k	3	43.42	2318	80.83	90.34	20.79
	gs1k	4	44.40	2829	81.58	90.82	20.06
	gs10k	1	35.92	1221	74.59	84.82	28.45
	gs10k	2	42.02	1862	78.16	88.17	22.07
	gs10k	3	43.34	2335	80.39	89.29	18.77
	gs10k	4	44.10	2837	79.94	88.63	17.81

Table 4: Full En-Zh runtime glossary-bank ablation. The runtime bank changes from the raw task glossary to GS-1k or GS-10k, while all terminology metrics use the corresponding fixed raw denominator. The main text plots the ACL rows; medicine rows are included as a stricter domain stress test.